Message-Level Communication Functions in LLM Multi-Agent Systems: A Taxonomy and Empirical Study

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Abstract

Removing the least informative messages from LLM multi-agent conversations produces the largest behavioral disruption. Across 2,693 GSM8K traces spanning three 7–9B backbones and three communication topologies, perturbation-based ablation reveals that excising simple acknowledgments changes downstream behavior far more (mean similarity $= 0.35$) than removing content-rich reasoning guidance (0.78); a single removed acknowledgment alters the final answer in 66% of cases versus 29% for reasoning guidance. We trace this pattern to three backbone-dependent communication modes: *function-driven* (Qwen; $F = 232$, $p < 0.001$), *position-driven* (Gemma; $R^2 = 0.34$, function type n.s.), and *undifferentiated* (Llama; $R^2 = 0.02$). A crossover interaction suggests that dynamic routing disrupts endogenous structure in function-driven models while partially activating it in undifferentiated ones, though mixed-effects analysis indicates this convergence is modulated by task-level clustering. Robustness checks across multiple similarity metrics, temperature settings, mixed-effects models, and a MATH cross-domain validation support these findings. These results question the assumptions underlying content-based pruning heuristics and suggest that communication structure in multi-agent systems is model-dependent rather than universal.

1 Introduction

Communication optimization in LLM-based multi-agent systems rests on an implicit assumption: message value correlates with information content. Pruning methods Wang et al. [2025b], Li et al. [2025], Zhang et al. [2025a] discard messages deemed redundant or low-content; routing methods Li et al. [2026], Chen et al. [2026a] direct information-rich messages to relevant agents. This assumption is intuitive—surely a detailed reasoning trace matters more than a brief "I agree"—but it has never been empirically tested at the message level.

We test it, and find the opposite. Through perturbation-based ablation of 2,693 multi-agent GSM8K traces across three backbones and three topologies, we show that removing informationally sparse acknowledgments produces the *largest* behavioral disruption (mean similarity $= 0.35$), while removing content-rich reasoning guidance produces the *smallest* (0.78). A single excised acknowledgment alters the final answer in 66% of cases, versus 29% for reasoning guidance. Content-based pruning heuristics would discard the wrong messages.

Why do low-content messages carry high impact? We identify three qualitatively distinct communication modes that explain this pattern:

1. **Impact gradient and structural load-bearing** (C1). Under fixed-turn topologies, the per-type impact gradient is inversely related to information content. A neutralize ablation—preserving struc-

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ture while removing content—produces the opposite gradient, confirming that acknowledgments' high impact stems from structural presence rather than informational content (Appendix E).

2. **Three backbone-dependent communication modes** (C2). OLS ANCOVA reveals that the impact gradient is not universal but backbone-dependent: Qwen exhibits *function-driven* communication, Gemma *position-driven*, and Llama *undifferentiated*. These are qualitatively distinct modes, not points on a gradient. C1's impact gradient is the instantiation of C2's function-driven mode on Qwen.
3. **Crossover interaction** (C3). Selector topology partially converges the function-driven and undifferentiated modes (backbone $\times$ topology $\eta^2 = 0.043$). Mixed-effects analysis reveals that Qwen's convergence is more thorough than OLS suggests: the function type effect vanishes entirely once task-level clustering is controlled (ICC $= 0.73$). This crossover is an instantiation of C2 under topological perturbation.

Cross-domain validation on MATH provides preliminary support for both the function-driven and position-driven modes; temperature robustness checks confirm stable results (Appendices H, C).

Our approach combines a five-category functional taxonomy (INFORMATION PROVISION, REASONING GUIDANCE, VERIFICATION, COORDINATION, ACKNOWLEDGMENT) validated via cross-model annotation ($\kappa = 0.35$–0.67) with perturbation-based ablation that isolates each message's downstream behavioral impact. Theoretical work Rizvi-Martel et al. [2026] establishes *when* communication helps; trace-level diagnosis Cemri et al. [2025] identifies *what went wrong*. We provide the missing *what*: a functional characterization of message-level contributions that suggests communication structure in multi-agent systems is model-dependent rather than universal, at least within the 7–9B scale and GSM8K domain studied here.

2 Related Work

Our work draws on four threads: theoretical foundations, LLM-based MAS, communication efficiency, and message function analysis.

Theoretical foundations. Rizvi-Martel *et al.* Rizvi-Martel et al. [2026] derive information-theoretic bounds on when multi-agent communication helps, establishing *when* communication is necessary but not *what* content the bandwidth should carry. The Five Ws survey Chen et al. [2026b] identifies "what to communicate" as the most understudied dimension. Our taxonomy directly fills this gap.

LLM-based MAS. Frameworks such as CAMEL Li et al. [2023], MetaGPT Hong et al. [2024], AG2 Wu et al. [2024], and AgentVerse Chen et al. [2024] demonstrate that inter-agent communication affects performance. Multi-agent debate Du et al. [2024] improves reasoning, though majority voting may largely account for the gains Choi et al. [2025]; controlled studies confirm that intrinsic reasoning strength dominates structural parameters Wu et al. [2025], and deliberation protocols can underperform simple best-of-N selection de Font-Reaulx et al. [2026]. Yan *et al.* Yan et al. [2025] provide a communication-centric survey organizing MAS research by architecture, strategy, and content, identifying message-level empirical analysis as an open direction. These studies establish that communication matters without characterizing the mechanisms through which it operates.

Communication efficiency and topology. AgentDropout Wang et al. [2025b], AgentPrune Zhang et al. [2025a], AGP Li et al. [2025], SupervisorAgent Lin et al. [2026], and OPTIMA Chen et al. [2025] achieve 20–90% token reduction by pruning messages, but treat message utility as a scalar without characterizing qualitative differences. ARG-Designer Li et al. [2026] and GoAgent Chen et al. [2026a] optimize topologies; information propagation studies Shen et al. [2025] use a binary correct/erroneous taxonomy. Our work provides the functional categories that efficiency and topology methods currently lack, and studies the backbone–topology interaction invisible to topology-only analyses.

Communication strategy analysis. S^2-MAD Liang et al. [2025] benchmarks multi-agent debate at scale; SHARP Kim et al. [2026] harmonizes agent roles for problem solving. Both evaluate system-level performance without message-level functional analysis. Most closely related to our work, Zhang *et al.* Zhang et al. [2025b] analyze dialogue acts in collaborative LLM agents on mathematical problem-solving, finding that peer-to-peer collaboration achieves the highest accuracy.

Table 1: Taxonomy of communication function types with definitions and examples from GSM8K solver–checker dialogues.

Function Type	Definition	Example
INFORMATION PROVISION	Supplies factual content, data, or intermediate results	"The total cost is 3 × $5 = $15 for the shirts."
REASONING GUIDANCE	Directs reasoning through decomposition or strategic suggestions	"First calculate the weekly cost, then multiply by 4."
VERIFICATION	Checks or corrects the correctness of prior messages	"17 × 3 = 51, not 53. The corrected answer is $51."
COORDINATION	Manages dialogue flow, assigns roles, or delegates tasks	"I'll solve first, then you verify step by step."
ACKNOWLEDGMENT	Simple confirmations or agreements with no new content	"I agree with your solution. The answer is 42."

Our work differs in three respects: (1) we measure *perturbation impact* through remove/neutralize ablation rather than descriptive dialogue act frequencies; (2) we provide statistical quantification via ANCOVA decomposition (F-statistics, R^2); and (3) we discover a backbone $\times$ topology crossover interaction invisible to single-backbone analyses.

Prompt sensitivity and pragmatics. LLM outputs are sensitive to prompt features Sclar et al. [2024], He et al. [2024]; our temperature robustness checks (Appendix C) partially control for this. Collaborative RSA Estienne et al. [2025] and TalkHier Wang et al. [2025a] model multi-turn pragmatics and hierarchical communication, providing complementary theoretical lenses.

Attribution and message function. AgentSHAP Horovicz et al. [2025] quantifies tool importance via Shapley values at the agent level. MAST Cemri et al. [2025] identifies 14 failure modes from 1,600+ traces but operates at the execution-trace level, not the message level. Our ablation framework provides finer-grained (individual message) and qualitatively richer (function type) analysis. Classical speech act theory Austin [1962], Searle [1969], FIPA ACL Foundation for Intelligent Physical Agents [2002], and dialogue act taxonomies Core and Allen [1997], Bunt et al. [2012] inform our categories, but our taxonomy is empirically grounded in LLM agent behavior and defined by measurable perturbation impact—a dimension absent from classical frameworks.

3 Taxonomy of Communication Functions

We define five exhaustive, mutually exclusive communication function types (Table 1). Ambiguity is resolved by hierarchical priority: *coordination* $\succ$ *verification* $\succ$ *reasoning_guidance* $\succ$ *information_provision* $\succ$ *acknowledgment*, reflecting the principle that structurally higher-level functions take precedence.

INFORMATION PROVISION supplies facts that expand the shared state without directing how they should be used. REASONING GUIDANCE is prescriptive: it tells the recipient *how* to proceed. VERIFICATION assesses prior contributions' correctness. COORDINATION manages meta-structure (roles, turn-taking) without contributing task content. ACKNOWLEDGMENT confirms receipt or agreement; while informationally sparse, it maintains conversational coherence.

Our categories map onto established dialogue act frameworks (DAMSL Core and Allen [1997], ISO 24617-2 Bunt et al. [2012], FIPA ACL Foundation for Intelligent Physical Agents [2002]; detailed mapping in Appendix L), collapsing 56+ fine-grained types into five categories operationalized via measurable perturbation impact rather than speaker intent.

Annotation protocol. Each message is annotated by an LLM annotator using a structured prompt containing taxonomy definitions, decision rules, and three in-context examples per type, with temperature 0 and constrained output. Cross-model validation uses two independent annotator models per message; we adopt $\kappa \geq 0.4$ as our target threshold for cross-model agreement and report Cohen's κ throughout (Section 6.4).

4 Methodology

4.1 Multi-Agent System Setup

We use AG2 v0.6.1 (AutoGen) Wu et al. [2024] with solver–checker agent teams across three backbones of comparable scale: Qwen2.5-7B-Instruct Yang et al. [2024], Llama-3.1-8B-Instruct Grattafiori et al. [2024], and Gemma-2-9B-IT Gemma Team [2024] (7–9B parameters). We study three topologies: **round-robin** (fixed cyclic order), **sequential** (strict two-phase rounds), and **selector** (dynamic routing via a selector agent). All models are served via vLLM v0.6.6 Kwon et al. [2023] with greedy decoding (temperature $= 0$, max tokens $= 2048$). For each of the $3 \times 3 = 9$ configurations, we collect traces on 300 GSM8K Cobbe et al. [2021] test-split tasks sampled with seed $= 42$, yielding 2,693 non-empty traces. The MATH cross-domain validation (Appendix H) uses 50 tasks at Level 3–5 (Algebra and Number Theory), also sampled with seed $= 42$. The full annotation prompt is reproduced in Appendix I.

4.2 Message-Level Ablation

To isolate each message's perturbation effect on downstream behavior, we introduce two ablation types. **Remove ablation** deletes message m_i from trace $\mathcal{T}$ and regenerates m_{i+1}. **Neutralize ablation** replaces m_i with "`Acknowledged.`", preserving positional structure while removing content. Comparing the two disentangles content effects from structural presence effects McNemar [1947].

We measure impact as:

$$\text{sim}(m_i) = \cos\big(\mathbf{e}(m_{i+1}),\, \mathbf{e}(m'_{i+1})\big) \tag{1}$$

where $\mathbf{e}(\cdot)$ is a sentence embedding (`all-MiniLM-L6-v2`; Reimers and Gurevych [2019]). Lower similarity indicates higher message impact. We report similarity throughout; impact $= 1 -$ sim.

4.3 Statistical Framework

We test whether function types produce different impact distributions using the Kruskal–Wallis H test Kruskal and Wallis [1952] (non-parametric, appropriate for bounded, skewed distributions), with Dunn's post-hoc tests Dunn [1964] and Bonferroni correction.

To control for the position confound—function types are non-uniformly distributed across conversation positions—we fit an OLS ANCOVA:

$$\text{sim}(m_i) = \beta_0 + \sum_{j=1}^{4} \beta_j \cdot \mathbb{1}[\text{type}(m_i) = j] + \gamma_1 \cdot \text{pos}(m_i) + \gamma_2 \cdot \text{round}(m_i) + \epsilon_i \tag{2}$$

The partial F-statistic for the function type factor tests whether function type effects persist after controlling for position and round number. We report HC3 heteroscedasticity-consistent standard errors throughout; residual diagnostics are in Appendix D. We further validate with a pooled three-way ANOVA across all backbones and topologies (Section 6.4).

5 Experiments

Each agent team uses AG2's `ConversableAgent` with task-specific system prompts (solver: step-by-step solutions; checker: verify and flag errors). Under selector topology, a third agent routes messages via `GroupChat` with `speaker_selection_method="auto"`. Conversations are limited to 10 rounds. Function type labels are assigned by two independent LLM annotators (Qwen2.5-7B as Model A, Llama-3.1-8B as Model B) with structured JSON output. Solution correctness is determined by comparing extracted numerical answers against GSM8K ground truth.

6 Results

We report results across all nine backbone–topology configurations.

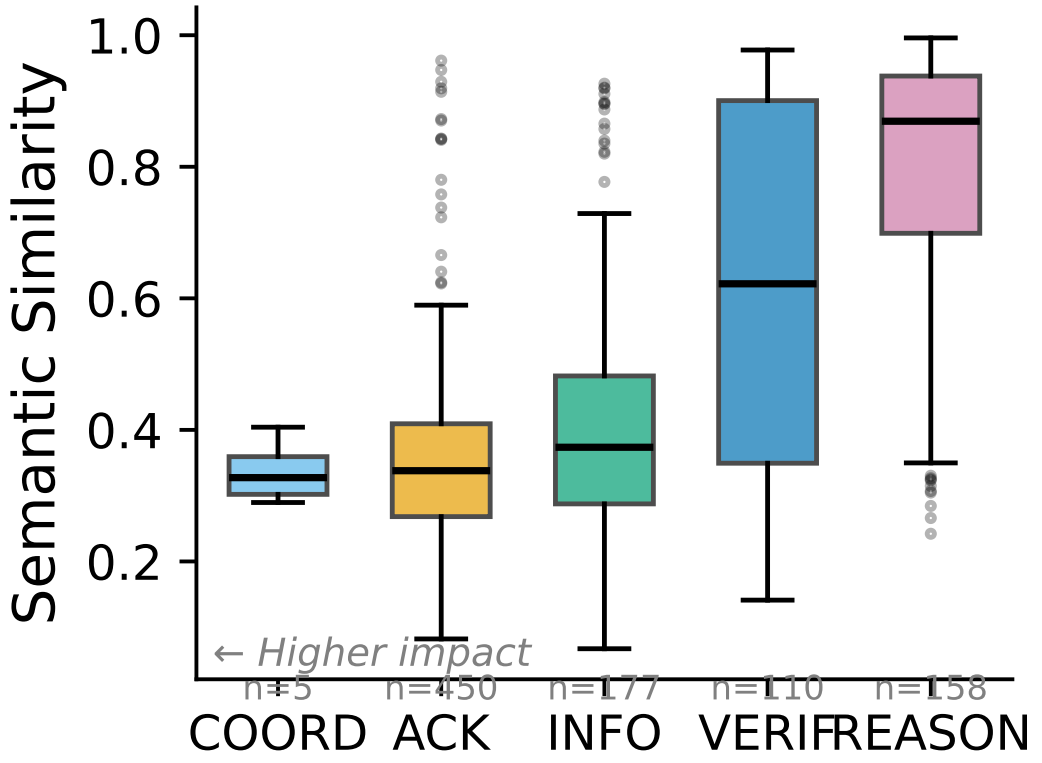


Figure 1: Remove-ablation semantic similarity (MiniLM cosine) by function type for Qwen2.5-7B under round-robin topology ($n = 895$). Lower similarity indicates greater behavioral disruption when the message is removed. Boxes show IQR; whiskers extend to $1.5\times$ IQR. Acknowledgment messages (median $= 0.35$) are the most impactful despite carrying the least information content.

6.1 Counter-Intuitive Impact Gradient

For Qwen under fixed-turn topologies (Figure 1), removing ACKNOWLEDGMENT messages—the least informative category—produces the largest behavioral change (mean similarity $= 0.352$, $n = 450$), followed by INFORMATION_PROVISION (0.410, $n = 177$), VERIFICATION (0.616, $n = 110$), and REASONING_GUIDANCE (0.778, $n = 158$). The impact gradient is inversely related to information content: informationally sparse acknowledgments are more impactful than content-rich reasoning guidance.

Pruning evidence. Individual-message ablation confirms this gradient at the single-message level: removing one acknowledgment alters the final answer in 66% of cases, versus 29% for reasoning guidance. The neutralize ablation—which preserves message structure while removing content—produces the opposite gradient: information-rich types suffer most. This double dissociation confirms that acknowledgment's high remove-impact stems from structural presence rather than informational content (Appendix E). The gradient is robust across four independent text similarity metrics (Spearman $\rho > 0.91$; Appendix G) and across RR and SEQ topologies, but breaks down under selector topology.

6.2 Three Communication Modes

The Kruskal–Wallis H test confirms that function type significantly predicts message impact across all configurations ($p < 0.01$), but the magnitude varies by an order of magnitude across backbones (Table 2). However, this omnibus test conflates function type effects with positional confounds.

Table 2: Kruskal–Wallis H statistics for semantic similarity across function types. All $p < 0.01$. Group counts vary (4–5 under RR/SEQ, 2–3 under selector); H values not directly comparable across topologies.

Backbone	RR	SEQ	Selector
Qwen2.5-7B	287.4	288.1	50.8
Gemma-2-9B	21.5	25.1	6.9
Llama-3.1-8B	13.0	15.8	63.8

Table 3: OLS ANCOVA: Type II partial F for function type (controlling for position and round) and full-model R^2. Type II SS is used because function type and position are not orthogonal; Type III SS yields identical conclusions (all significance levels unchanged). $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; n.s. = not significant.

	RR		SEQ		Selector	
Backbone	F	R^2	F	R^2	F	R^2
Qwen2.5-7B	232^{***}	.450	225^{***}	.446	55.8^{***}	.065
Gemma-2-9B	$0.43^{\text{n.s.}}$	.341	$0.44^{\text{n.s.}}$	.333	$0.60^{\text{n.s.}}$	.036
Llama-3.1-8B	3.81^{*}	.021	4.45^{**}	.020	17.3^{***}	.056

Position control via OLS ANCOVA. To disentangle function type from positional confounds, we fit the ANCOVA model (Equation 2). Table 3 reveals three qualitatively distinct communication modes that explain *why* the impact gradient exists for some backbones but not others.

- **Function-driven** (Qwen): function type is the dominant predictor ($F = 232$), driving most of the $R^2 = 0.45$. The impact gradient (§6.1) is a direct consequence: because function type determines impact, removing an entire functional category produces category-specific behavioral changes.
- **Position-driven** (Gemma): $R^2 = 0.34$ but function type $F = 0.43$ (n.s.)—all explained variance comes from message position and round number. Adding a quadratic position term absorbs $\Delta R^2 = 0.228$, rendering function type non-significant ($p = 0.806$; Appendix D), confirming that Gemma's impact variation is driven entirely by nonlinear positional dynamics.
- **Undifferentiated** (Llama): $R^2 = 0.02$; function type is statistically significant ($F = 3.81$, $p = .01$; confirmed under mixed-effects, $p = .002$) but with practically negligible effect size (marginal $R^2 < 0.03$). Neither function type nor position explains substantial variance.

Figure 2 shows the function type distribution across backbones under round-robin topology.

6.3 Backbone × Topology Crossover

Figure 3 visualizes the crossover interaction. Qwen is robustly function-driven under RR and SEQ (ME LRT $\chi^2 > 150$, $p < .001$), but under selector its OLS F drops from 232 to 55.8 (R^2: $0.45 \rightarrow 0.065$) as its distribution collapses to two types (ACKNOWLEDGMENT $n = 324$, VERIFICATION $n = 567$); a mixed-effects model renders this residual effect non-significant (LRT $\chi^2 = 1.45$, $p = .229$; ICC $= 0.73$), indicating that selector-induced task-level clustering fully absorbs the function type signal. Llama's F rises from 3.81 to 17.3 (R^2: $0.02 \rightarrow 0.056$) as it expands to three types (COORDINATION $n = 220$, VERIFICATION $n = 217$, ACKNOWLEDGMENT $n = 428$); this effect is confirmed by mixed-effects (LRT $\chi^2 = 22.5$, $p < .001$). Gemma's R^2 drops ($0.34 \rightarrow 0.036$) due to disruption of positional regularity rather than functional collapse, since function type remains non-significant under both OLS ($F = 0.60$) and mixed-effects for RR/SEQ. The mixed-effects analysis reveals that selector's disruption of Qwen is more thorough than OLS suggests: not only does R^2 drop, but the function type effect itself vanishes once task-level variance is accounted for.

6.4 Statistical Validation

Pooled three-way ANOVA. A three-way ANOVA on all $N = 8{,}065$ messages (Table 4) confirms backbone as the dominant factor ($\eta^2 = 0.135$, $F = 629$) and the crossover interaction (backbone × topology $\eta^2 = 0.043$, $F = 181$). The function × backbone interaction ($\eta^2 = 0.063$) is 12× larger

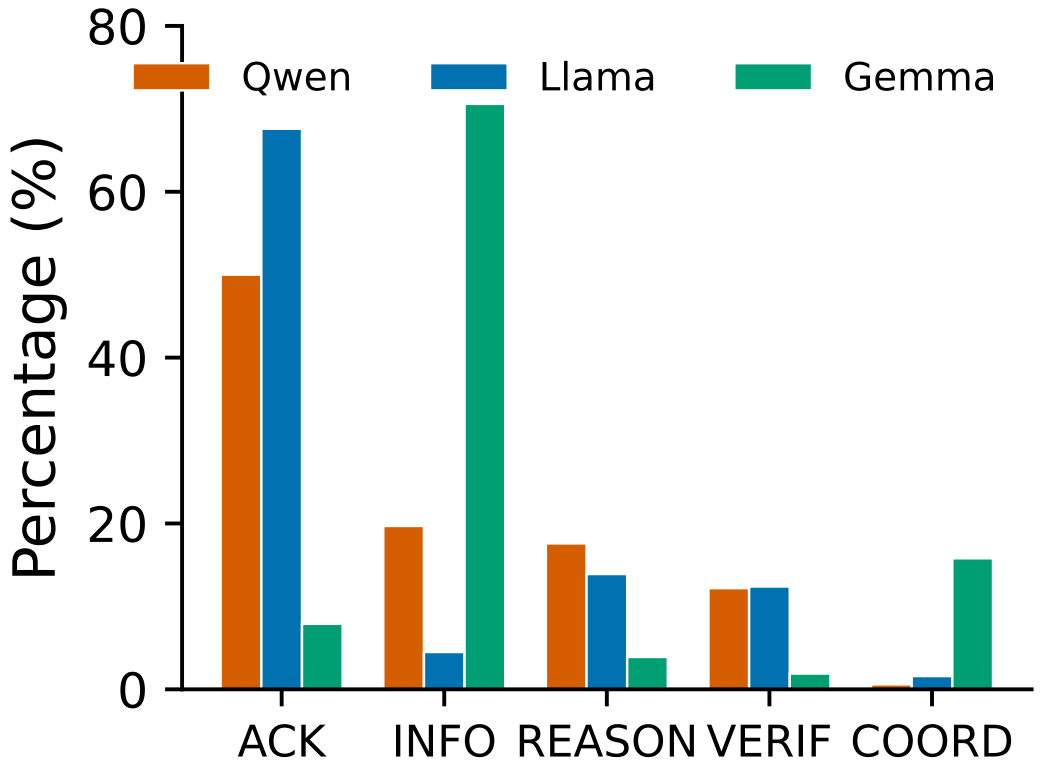


Figure 2: Function type distribution by backbone under round-robin topology (300 GSM8K tasks each). Qwen2.5-7B and Llama-3.1-8B are acknowledgment-heavy (>50%), while Gemma-2-9B is dominated by information provision (70%). Types with $n < 30$ are excluded from OLS ANCOVA analysis.

Table 4: Pooled ANOVA on remove-ablation similarity ($N = 8{,}065$, $R^2 = 0.298$). All $p < .001$.

Term	η^2	F
Backbone	0.135	628.7
Function × Backbone	0.063	67.1
Topology class	0.051	435.4
Backbone × Topology	0.043	181.1
Three-way	0.021	21.1
Function × Topology	0.014	27.7
Function type	0.005	10.4

than the function main effect ($\eta^2 = 0.005$), reflecting that function type is a strong predictor for Qwen but negligible for Gemma and Llama—pooling dilutes the main effect while preserving the interaction.

Mixed-effects robustness. To address within-trace message correlation, we fit linear mixed-effects models with task as a random intercept and the same fixed effects as the OLS ANCOVA (function type, normalized message position, round number). Seven of nine configurations yield the same significance conclusion under mixed-effects as under OLS (Appendix J). The two exceptions both occur under selector topology: Qwen SEL shifts from significant (OLS $F = 55.8$, $p < .001$) to non-significant (LRT $\chi^2 = 1.45$, $p = .229$) with ICC $= 0.73$, indicating that task-level clustering subsumes the OLS function type effect; Gemma SEL shifts in the opposite direction, from non-significant (OLS $F = 0.60$) to significant (LRT $\chi^2 = 14.8$, $p < .001$, ICC $= 0.33$), though this comparison involves only two function types (1 df) after excluding three low-frequency categories, making the result suggestive but requiring larger samples to confirm. Selector topology introduces

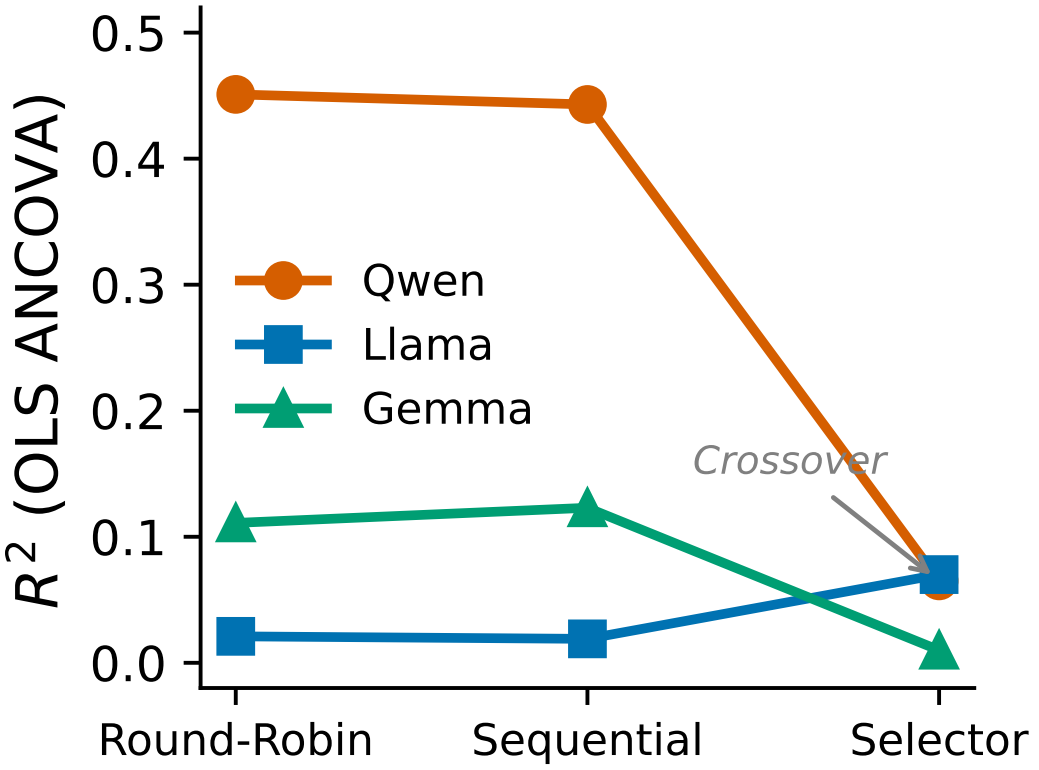


Figure 3: Backbone × topology crossover interaction showing full-model R^2 from OLS ANCOVA (Equation 2). Higher R^2 indicates more variance explained by the model (function type + position). Qwen's functional structure ($R^2 = 0.45$) degrades under selector routing ($R^2 = 0.065$) while Llama's activates ($0.02 \rightarrow 0.056$), partially converging. Gemma's position-driven R^2 collapses from 0.34 to 0.04 as positional regularity is disrupted.

Table 5: GSM8K accuracy (%) by backbone and topology.

Backbone	RR	SEQ	Selector
Qwen2.5-7B	89.7	90.0	89.7
Gemma-2-9B	80.3	80.7	83.0
Llama-3.1-8B	74.8	71.6	56.2

strong task-level clustering (ICC > 0.3 for all backbones except Gemma RR/SEQ), making OLS unreliable for this configuration. Llama is statistically significant under mixed-effects across all topologies ($p < .003$) but with practically negligible effect sizes (marginal $R^2 < 0.03$), consistent with its classification as undifferentiated.

Cross-model annotation agreement. Cross-model κ ranges from 0.35 (Gemma) to 0.67 (Qwen), spanning fair-to-substantial agreement Landis and Koch [1977]. Qwen's highest κ is consistent with function-driven communication being easier to classify. Gemma falls below our target threshold ($\kappa = 0.35$), indicating weaker cross-model consistency for this backbone. This aligns with its position-driven mode: if function type is not the organizing principle, annotators naturally disagree more on function labels. Per-type κ reveals a complementary pattern: Qwen shows high VERIFICATION agreement ($\kappa \approx 0.66$) but low INFORMATION_PROVISION ($\kappa \approx 0.08$); Gemma shows the reverse ($\kappa \approx 0.87$ and 0.04, respectively). Full agreement tables are in Appendix F.

Task accuracy and surface vs. effective differentiation. Qwen's accuracy is topology-invariant ($\sim$90%) despite dramatic communication changes under selector. Llama presents a paradox: functional differentiation *increases* under selector yet accuracy *drops* ($74.8\% \rightarrow 56.2\%$). This dissociates

surface differentiation (externally induced diversity) from *effective differentiation* (endogenous specialization that aids task performance). Among Llama's COORDINATION messages under selector, only 7.9% are correct (vs. 83.3% for VERIFICATION); tasks with coordination messages achieve 21.6% success vs. 54.5% without.

7 Discussion

Conversational coherence as structural scaffold. The counter-intuitive impact gradient admits a *conversational coherence* explanation: acknowledgments are informationally sparse but contextually load-bearing, and their absence violates conversational expectations. The neutralize ablation corroborates this—neutralizing has the opposite gradient (information-rich types suffer most), confirming that acknowledgments' high remove-impact stems from structural presence rather than content (Appendix E). The neutralize replacement text ("Acknowledged.") is itself acknowledgment-like; a three-placeholder sensitivity test confirms the neutralize gradient is robust across replacement choices ($\rho = 0.90$ between content-neutral placeholders; Appendix M), and the remove ablation (primary result) is unaffected since it deletes messages entirely.

Three communication modes, not a gradient. The ANCOVA decomposition (Table 3) reveals that the impact gradient is not universal but depends on backbone-specific communication modes: *function-driven* (Qwen), *position-driven* (Gemma), and *undifferentiated* (Llama). We hypothesize this reflects instruction-following fidelity—models that more precisely modulate output produce messages whose functional roles are behaviorally distinct. A fourth backbone (Mistral-7B; Appendix K) shows a *weakly differentiated*, topology-sensitive mode. Task difficulty stratification confirms all three modes persist on hard tasks (Appendix N), ruling out a capability artifact.

Practical implications. First, systems should evaluate message prunability through perturbation probing rather than information-content heuristics. Second, optimal topology depends on communication mode: dynamic routing benefits undifferentiated communicators but disrupts function-driven ones (§6.3); mixed-effects analysis confirms that selector introduces task-level dependencies altering significance conclusions (Appendix J). Third, communication diversity alone is insufficient—Llama's selector paradox (more diverse, lower accuracy) shows evaluation must couple functional analysis with task performance.

Limitations and alternative explanations. The impact gradient could reflect *pragmatic alignment* rather than structural scaffolding, or backbone-dependent modes could arise from *context distributional shift*; definitive tests require probing model internals. Our primary experiments use GSM8K; cross-domain pilots (Appendices B, H) suggest applicability but with reduced effect sizes. Three backbones at 7–9B provide architectural diversity but no scale variation. Mixed-effects models confirm 7/9 OLS conclusions; the two exceptions under selector reflect high task-level clustering (Appendix J). LLM annotation creates potential circularity mitigated by cross-model validation ($\kappa = 0.35$–0.67); core findings rest on Qwen ($\kappa = 0.67$) and Llama ($\kappa = 0.46$–0.63). All experiments use a single framework and prompt template; prompt sensitivity Sclar et al. [2024] could shift distributions, though temperature checks suggest stability (Appendix C).

8 Conclusion

Perturbation-based ablation across 2,693 multi-agent traces reveals that informationally sparse acknowledgments are the most behaviorally load-bearing messages in LLM multi-agent systems. This gradient is explained by three backbone-dependent communication modes—function-driven, position-driven, and undifferentiated—whose interaction with topology implies that optimal routing depends on communication mode. These findings question content-based pruning assumptions and suggest that communication structure is an emergent, model-dependent property.

References

John L. Austin. *How to Do Things with Words*. Oxford University Press, 1962.

Harry Bunt, Jan Alexandersson, Jae-Woong Choe, Alex Chengyu Fang, Koiti Hasida, Volha Petukhova, Andrei Popescu-Belis, and David R. Traum. Towards an ISO standard for dialogue act annotation. *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC)*, 2012.

Mert Cemri et al. Why do multi-agent LLM systems fail? *arXiv preprint arXiv:2503.13657*, 2025.

Hongjiang Chen et al. GoAgent: Group-optimized multi-agent topology for efficient communication. *arXiv preprint arXiv:2603.19677*, 2026a.

Jingdi Chen et al. The five ws of multi-agent communication. *Transactions on Machine Learning Research (TMLR)*, 2026b. arXiv:2602.11583.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Weize Chen, Yusheng Su, Jingwei Zuo, Cheng Yang, Chenfei Yuan, Chi-Min Chan, Heyang Yu, Yaxi Lu, Yi-Hsin Hung, Chen Qian, et al. AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors. In *ICLR*, 2024.

Weize Chen, Jiarui Yuan, Chen Qian, et al. OPTIMA: Optimizing effectiveness and efficiency for LLM-based multi-agent system. In *ACL Findings*, 2025. arXiv:2410.08115.

Hyeong Kyu Choi, Xiaojin Zhu, and Yixuan Li. Debate or vote: Which yields better decisions in multi-agent large language models? In *NeurIPS*, 2025. Spotlight. arXiv:2508.17536.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Mark G. Core and James F. Allen. Coding dialogs with the DAMSL annotation scheme. In *AAAI Fall Symposium on Communicative Action in Humans and Machines*, 1997.

Paul de Font-Reaulx et al. DeliberationBench: When do more voices hurt? a controlled study of multi-LLM deliberation protocols. *arXiv preprint arXiv:2601.08835*, 2026.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *ICML*, 2024.

Olive Jean Dunn. Multiple comparisons using rank sums. *Technometrics*, 6(3):241–252, 1964.

Lautaro Estienne, Gabriel Ben Zenou, Nona Naderi, Jackie CK Cheung, and Pablo Piantanida. Collaborative rational speech act: Pragmatic reasoning for multi-turn dialog. In *EMNLP*, pages 22520–22534, 2025. arXiv:2507.14063.

Foundation for Intelligent Physical Agents. FIPA communicative act library specification. Technical report, FIPA, 2002. Document SC00037J.

Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Jia He, Mukund Rungta, David Koleczek, Arshdeep Sekhon, Franklin X Wang, and Sadid Hasan. Does prompt formatting have any impact on LLM performance? *arXiv preprint arXiv:2411.10541*, 2024.

Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, et al. MetaGPT: Meta programming for a multi-agent collaborative framework. In *ICLR*, 2024.

Miriam Horovicz et al. AgentSHAP: Quantifying tool importance in LLM agents via shapley values. *arXiv preprint arXiv:2512.12597*, 2025.

Hyunwoo Kim, Jinhyuk Park, Sangwoo Lee, and Kyunghyun Cho. SHARP: Harmonizing agent roles in multi-LLM problem solving. *arXiv preprint arXiv:2602.08335*, 2026.

William H. Kruskal and W. Allen Wallis. Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260):583–621, 1952.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *SOSP*, 2023.

J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977.

Boyi Li, Zhonghan Zhao, Der-Horng Lee, and Gaoang Wang. Adaptive graph pruning for multi-agent communication. In *ECAI*, 2025. arXiv:2506.02951.

Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for "mind" exploration of large language model society. *NeurIPS*, 2023.

Shiyuan Li et al. Assemble your crew: Automatic multi-agent communication topology design via autoregressive graph generation. In *AAAI*, 2026. Oral. arXiv:2507.18224.

Zhenyang Liang, Luyi Shi, Qisen Wu, Huajie Yuan, Shuangyin Chen, and Guanqi Li. S^2-MAD: A scalable and standardized framework for benchmarking multi-agent debate. *arXiv preprint arXiv:2502.04790*, 2025.

Fulin Lin et al. Stop wasting your tokens: SupervisorAgent as an LLM-free adaptive filter at interaction points. In *ICLR*, 2026. arXiv:2510.26585.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

Vincent Rizvi-Martel et al. Benefits and limitations of communication in multi-agent reasoning. In *ICLR*, 2026. arXiv:2510.13903.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *ICLR*, 2024. arXiv:2310.11324.

John R. Searle. *Speech Acts: An Essay in the Philosophy of Language*. Cambridge University Press, 1969.

Xu Shen, Yixin Liu, Yiwei Dai, et al. Understanding the information propagation effects of communication topologies in LLM-based multi-agent systems. In *EMNLP*, 2025. arXiv:2505.23352.

Zhao Wang, Sota Moriyama, Wei-Yao Wang, Briti Gangopadhyay, and Shingo Takamatsu. Talk structurally, act hierarchically: A collaborative framework for LLM multi-agent systems. *arXiv preprint arXiv:2502.11098*, 2025a.

Zhilong Wang et al. AgentDropout: Dynamic agent elimination for token-efficient and high-performance LLM-based multi-agent collaboration. *arXiv preprint arXiv:2503.18891*, 2025b.

Haolun Wu, Zhenkun Li, and Lingyao Li. Can LLM agents really debate? a controlled study of multi-agent debate in logical reasoning. *arXiv preprint arXiv:2511.07784*, 2025.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. In *COLM*, 2024.

Bingyu Yan, Zhibo Zhou, Litian Zhang, Lian Zhang, Ziyi Zhou, Dezhuang Miao, Zhoujun Li, Chaozhuo Li, and Xiaoming Zhang. Beyond self-talk: A communication-centric survey of LLM-based multi-agent systems. *arXiv preprint arXiv:2502.14321*, 2025.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5: A party of foundation models. *arXiv preprint arXiv:2412.15115*, 2024.

Guibin Zhang, Yanwei Yue, et al. Cut the crap: An economical communication pipeline for LLM-based multi-agent systems. In *ICLR*, 2025a. arXiv:2410.02506.

Liang Zhang, Xiaoming Zhai, Jionghao Lin, Jennifer Kleiman, Diego Zapata-Rivera, Carol Forsyth, Yang Jiang, Xiangen Hu, and Arthur C. Graesser. Exploring communication strategies for collaborative LLM agents in mathematical problem-solving. In *AIED*, 2025b. arXiv:2507.17753.

A Qualitative Examples of Communication Functions

Table 6 presents representative messages for each of the five communication function types identified in our taxonomy, selected from annotations with full inter-annotator agreement and maximum confidence scores. Examples are drawn from different backbone–topology combinations to illustrate both the consistency of function definitions and the stylistic variations across models.

B HumanEval Cross-Domain Pilot

To provide preliminary evidence of cross-domain applicability, we ran a 50-task pilot using Qwen2.5-7B with a hierarchical (solver–reviewer) topology on the HumanEval code generation benchmark Chen et al. [2021].

Results. The system achieved 80% pass@1 accuracy (40/50 tasks). The communication function distribution differs markedly from GSM8K: REASONING_GUIDANCE dominates (50.8%), followed by VERIFICATION (41.5%), with ACKNOWLEDGMENT (4.2%), COORDINATION (1.7%), and INFORMATION_PROVISION (1.7%) appearing rarely. Cross-model annotation agreement reached $\kappa = 0.47$ (moderate), comparable to the lower end of the GSM8K range.

Limitations. The hierarchical topology proved too efficient for statistical analysis: 86% of tasks were solved in a single round (2 messages), yielding only 118 total messages and 18 eligible for local impact analysis. Most function types had $n < 30$, precluding Kruskal–Wallis and OLS ANCOVA tests. The distributional shift (from acknowledgment-heavy in GSM8K to reasoning-guidance-heavy in HumanEval) is itself informative—it suggests that the taxonomy captures domain-specific communication patterns—but the statistical framework requires substantially more multi-round conversations to produce meaningful results in this domain.

C Temperature Robustness

To assess sensitivity to decoding temperature, we ran a 50-task subset of Qwen $\times$ round-robin at temperature $= 0.7$ (primary experiments use greedy decoding). Accuracy remains stable (90.0% vs. 89.7%). The Kruskal–Wallis test remains significant ($H = 34.0$, $p < 0.001$), and OLS ANCOVA confirms function-driven communication ($F = 37.9$, $p < 0.001$; $R^2 = 0.50$; cross-model $\kappa = 0.60$).

[1]This Llama message was classified as REASONING GUIDANCE by both annotators with full agreement, despite its surface resemblance to ACKNOWLEDGMENT. This misclassification illustrates Llama's undifferentiated communication mode (Section 7): its messages are functionally interchangeable regardless of nominal category, making even high-confidence annotations unreliable for this backbone.

Table 6: Qualitative examples for each communication function type. All examples have inter-annotator agreement with confidence $= 1.0$. Messages are lightly edited for length.

Function Type	Configuration	Message Excerpt
REASONING GUIDANCE (2,327 agreed)	Qwen / RR	*"To solve this problem, we need to convert the length of the wire from feet to inches and then divide by the length of each piece. 1. Convert 4 feet to inches: Since 1 foot = 12 inches, 4 feet* $= 4 \times 12 = 48$ *inches. 2. Divide the total length in inches by the length of each piece..."*
	Gemma / Selector	*"Here's how to solve the problem step-by-step: Start: 2 apples. First throw:* $2 + 5 = 7$ *apples. Second throw:* $7 - 1 + 6 = 12$ *apples. The answer is 12."*
	Llama / RR	*"I'm glad the solution was accurate. If you have another problem you'd like me to check, feel free to ask."*[1]
VERIFICATION (1,001 agreed)	Qwen / RR	*"Step 1: Calculate the initial amount of money Johnny has. Johnny's allowance: \$20. Extra money added: \$10. Total initial amount: \$20 + \$10 = \$30. Step 2: Determine the amount after the investment triples. Tripling the initial amount: \$30 × 3 = \$90. The answer is 90."*
	Llama / RR	*"The given problem is solved step by step. The final answer is 36."*
	Gemma / Selector	*"Here's how to solve the problem: Find the total cost: \$12 (adult) + \$10 (child) = \$22. Add the change: \$22 + \$8 = \$30. Answer: Brittany's mom gave the cashier \$30."*
ACKNOWLEDGMENT (1,854 agreed)	Qwen / RR	*"That is correct. Each single bagel will cost \$1.00 after the discount. The answer is 1."*
	Llama / RR	*"You've correctly summarized the steps and confirmed the calculations. The sequence of events and the final calculation are accurate. The answer is 43."*
	Gemma / RR	*"Your solution is perfectly correct! You clearly laid out the steps and used algebra effectively to find the answer. The answer is 10."*
INFORMATION PROVISION (471 agreed)	Gemma / RR	*"Here's the solution: Sunday's movies: They watch 4 movies / 2 = 2 movies on Sunday. Total weekend movies: 4 + 2 = 6 movies each weekend. Movies in 4 weeks: 6 × 4 = 24 movies. The answer is 24."*
	Qwen / RR	*"To determine how much crepe paper James needs, we first calculate the perimeter of his rectangular backyard fence..."*
	Llama / RR	*"Here's another problem: A bakery is making a special batch of cookies. They need to package 480 cookies into boxes that hold 12 cookies each. How many boxes can they fill with the 480 cookies? Let's solve this problem step by step."*
COORDINATION (595 agreed)	Llama / RR	*"I'm glad I was able to accurately solve the problem. If you have any more math problems you'd like me to solve, feel free to ask."*
	Gemma / RR	*"Please provide me with the problem you'd like me to solve! I'm ready to check the solver's solution..."*
	Llama / Selector	*"I'm glad you were able to provide the correct steps and solution. You have a good understanding of the problem..."*

A parallel check on Llama × round-robin at temperature $= 0.7$ yields $R^2 = 0.10$, $\kappa = 0.41$, consistent with its undifferentiated mode under greedy decoding. The function distribution shifts toward more REASONING_GUIDANCE (37% vs. 18%) and less ACKNOWLEDGMENT (36% vs. 50%), suggesting stochastic decoding produces more substantive content without altering the functional structure.

D OLS Robustness Checks

We validate the OLS ANCOVA (Equation 2) with three robustness checks across all nine configurations.

Residual diagnostics. Shapiro–Wilk tests reject normality for all configurations (expected at $n > 400$); Breusch–Pagan tests detect heteroscedasticity in all cases. We therefore report HC3 robust standard errors throughout the main text. All conclusions are unchanged under robust inference. VIF analysis shows high collinearity between `message_position` and `round_number` (VIF > 10), but function type dummy VIF values remain below 3, confirming that multicollinearity does not affect the key func_type inference. Q-Q plots are available in the supplementary materials.

Nonlinear position model. We augment the linear model with a quadratic position term (`position`^{2}). Table 7 reports key comparisons. For Qwen and Llama, adding `position`^{2} produces negligible $\Delta R^2 < 0.005$ and function type remains significant ($p < 0.001$). For Gemma, `position`^{2} absorbs $\Delta R^2 = 0.228$ (RR) and 0.215 (SEQ), rendering function type non-significant ($p > 0.69$). This confirms that Gemma's position-driven mode reflects genuine nonlinear positional dynamics rather than a linear-model artifact.

Table 7: Linear vs. quadratic position model: function type significance after controlling for nonlinear position effects.

Configuration	**Linear** R^2	**Quad** R^2	ΔR^2	**func_type** p **(quad)**
Qwen RR	.451	.452	.001	$< .001$
Qwen SEQ	.443	.447	.004	$< .001$
Qwen SEL	.065	.065	.000	$< .001$
Gemma RR	.111	.339	**.228**	.806
Gemma SEQ	.123	.338	**.215**	.699
Gemma SEL	.010	.040	.031	.219
Llama RR	.021	.022	.001	.010
Llama SEQ	.019	.020	.001	.004
Llama SEL	.070	.074	.004	$< .001$

Message length control. Adding message length (character count) as a covariate produces $\Delta R^2 < 0.04$ for all configurations. Function type remains significant for Qwen ($p < 0.001$) and Llama ($p < 0.03$) after controlling for length, confirming that the impact gradient is not an artifact of message length differences across function types.

E Neutralize Ablation Analysis

While the main text reports remove-ablation impact, neutralize ablation (replacing a message with "`Acknowledged.`" while preserving its slot) measures a complementary dimension. The two ablation types produce *opposite* impact gradients:

- **Remove** (content + structure removed): acknowledgment most impactful (sim $= 0.35$), reasoning guidance least (0.78).
- **Neutralize** (content removed, structure preserved): information provision most impactful (sim $= 0.05$), verification least (0.50).

The difference $\Delta = \text{sim}_{\text{remove}} - \text{sim}_{\text{neutralize}}$ isolates the *structural value* of each message type. Information provision shows the largest $\Delta = 0.47$ (almost entirely content value), while coordination shows $\Delta = 0.07$ (almost entirely structural value). This decomposition confirms that

acknowledgment's high remove-impact is driven by structural presence—violating conversational expectations—rather than informational content.

F Supplementary Tables

Table 8 reports the full cross-model annotation agreement matrix.

Table 8: Cross-model annotation agreement (Cohen's κ) by backbone and topology.

Backbone	Round-Robin	Sequential	Selector
Qwen2.5-7B	0.64	0.64	0.669
Gemma-2-9B	0.350	0.348	0.351
Llama-3.1-8B	0.46	0.47	0.634

G Impact Metric Robustness

To verify that the impact gradient (Section 6.3) is not an artifact of the MiniLM embedding metric, we recompute message impact using three independent text similarity measures: ROUGE-L, token-level F1, and Jaccard similarity. Table 9 reports the Spearman rank correlation (ρ) between MiniLM cosine similarity and each alternative metric across all nine configurations.

Table 9: Spearman ρ between MiniLM cosine similarity and alternative impact metrics. All correlations computed over per-message impact scores within each configuration.

Configuration	ROUGE-L	Token F1	Jaccard
Qwen RR	1.0	1.0	1.0
Qwen SEQ	1.0	1.0	1.0
Qwen SEL	1.0	1.0	1.0
Gemma RR	1.0	1.0	1.0
Gemma SEQ	1.0	0.8	0.8
Gemma SEL	0.949	0.949	0.949
Llama RR	0.6	0.6	0.6
Llama SEQ	1.0	1.0	1.0
Llama SEL	0.9	0.9	0.9
Mean	**0.939**	**0.917**	**0.917**

The prevalence of $\rho = 1.0$ reflects a ceiling effect: with only 4–5 function types per configuration, Spearman rank correlation over group medians has limited resolution—any two metrics that preserve the same ordinal ranking produce $\rho = 1.0$ regardless of magnitude differences. Where rankings do diverge (Llama RR $\rho = 0.6$, Gemma SEQ $\rho = 0.8$), the disagreement involves adjacent categories with similar impact scores, not reversals of the overall gradient. The mean $\rho > 0.91$ across all metric pairs confirms that the impact gradient ordering is consistent across metrics, though the ceiling effect limits the strength of this claim for individual configurations.

H Cross-Domain Validation: MATH

To test whether the three communication modes generalize beyond elementary arithmetic, we ran the full pipeline on 50 MATH tasks (Level 3–5, Algebra and Number Theory) using the same round-robin topology.

Qwen on MATH. Accuracy is 72.0% (36/50), lower than GSM8K (89.7%) as expected for harder tasks. Each task produces 6 messages (300 total, 250 eligible for analysis after excluding initial prompts). Cross-model annotation agreement reaches $\kappa = 0.45$ (moderate, above threshold). The OLS ANCOVA yields $R^2 = 0.098$ with function type significant after position control ($F = 3.77$, $p = 0.011$), confirming that Qwen's function-driven mode persists on a harder mathematical domain, though with reduced effect size ($R^2 = 0.098$ vs. 0.45 on GSM8K). The Kruskal–Wallis test is borderline ($H = 7.37$, $p = 0.061$), likely due to reduced sample size.

Gemma on MATH. Accuracy drops to 40.0% (20/50). Cross-model $\kappa = 0.48$ (above threshold). The OLS ANCOVA yields $R^2 = 0.236$ but function type is non-significant ($F = 0.09$, $p = 0.76$), while message position is highly significant ($F = 15.22$, $p < 0.001$). This replicates Gemma's position-driven mode from GSM8K: on MATH, position again explains the variance while function type does not. The Kruskal–Wallis test is also non-significant ($H = 2.05$, $p = 0.15$).

Llama on MATH. Accuracy is 64.3% (27/42; 8 tasks produced empty traces). Cross-model $\kappa = 0.36$ (below threshold). With only 168 total messages, ACKNOWLEDGMENT dominates (56%), and most other function types fall below $n = 30$, precluding OLS ANCOVA and Kruskal–Wallis analysis. The acknowledgment-dominated distribution is directionally consistent with Llama's undifferentiated mode on GSM8K, though statistical confirmation requires larger samples.

Summary. Qwen's function-driven mode ($F = 3.77$, $p = 0.011$) and Gemma's position-driven mode ($F = 0.09$, n.s.; position $F = 15.22$, $p < 0.001$) both replicate on MATH. Llama's distribution is directionally consistent with its undifferentiated mode but lacks statistical power. Two of three communication modes are confirmed to generalize beyond elementary arithmetic.

I Annotation Prompt

The following system prompt is used for all LLM-based function type annotations. Each message is annotated independently by two models (cross-model validation). Temperature is set to 0 with constrained JSON output.

```
You are a communication analyst for multi-agent systems.  Your task
is to classify the function of a message in a multi-agent math
problem-solving conversation.

Function Types

1.  information_provision:  Provides facts, data, intermediate
calculations, or partial results that advance the solution.
Includes reporting computed values, stating known quantities, or
presenting new numerical results.

2.  reasoning_guidance:  Provides reasoning direction, problem
decomposition, step-by-step plans, or conceptual frameworks.
Includes breaking the problem into sub-problems, suggesting solution
strategies, or outlining multi-step approaches.

3.  verification:  Checks, corrects, or confirms the correctness of
previous work.  Includes identifying errors, validating results,
pointing out mistakes, or confirming a step is correct with a
specific reason.

4.  coordination:  Manages the flow of the conversation, assigns
tasks, requests clarification, or controls turn-taking.  Includes
delegating sub-tasks, asking for more information, or directing the
problem-solving process.

5.  acknowledgment:  Simple confirmation, repetition, or agreement
that adds no new information or reasoning.  Includes restating what
was already said without extending it.

Decision Rules

- If a message both provides new results AND checks previous work,
classify as verification.
- If a message provides a reasoning plan AND new results, classify as
reasoning_guidance.
- If a message confirms correctness WITH a substantive explanation,
classify as verification.
- If a message simply agrees or restates without adding new
information, classify as acknowledgment.
```

```
Output Format

Respond with ONLY a JSON object:
{''type'': ''<function_type>'', ''confidence'': <0.0-1.0>, ''reasoning'':
''<brief explanation>''}
```

The user prompt provides the conversation context with all messages and marks the target message for classification:

```
## Conversation Context
[Message 0] {sender} -> {receiver}:
{content}
[Message 1] {sender} -> {receiver}: «< TARGET MESSAGE »>
{content}
[Message 2] {sender} -> {receiver}:
{content}
...
Classify the function of Message {target_idx}.
```

J Mixed-Effects Model Results

To address within-trace non-independence (multiple messages per task), we fit linear mixed-effects models with task as a random intercept and the same fixed effects as the OLS ANCOVA (function type, normalized message position, round number). Model specification was verified by reproducing the original OLS results (e.g., Gemma RR: $F = 0.43$, $R^2 = 0.341$, matching Table 3). Table 10 compares OLS and mixed-effects results. The likelihood ratio test (LRT) compares the full model against a reduced model without function type dummies.

Table 10: OLS ANCOVA vs. mixed-effects model (ME) with task random intercept. LRT = likelihood ratio test for function type. ICC = intraclass correlation from the full ME model. "Excluded" lists function types with $n < 30$ removed before fitting. $^{***}p < 0.001$; $^{**}p < 0.01$; n.s. = not significant. F and χ^2 values are not directly comparable across configurations due to differing numbers of included function types.

Config	n	OLS F	LRT χ^2	LRT p	ICC	Excluded
Qwen RR	895	232^{***}	156^{***}	$<.001$	.312	coord.
Qwen SEQ	894	225^{***}	166^{***}	$<.001$	.266	coord., unk.
Qwen SEL	891	55.8^{***}	$1.45^{\text{n.s.}}$	.229	**.726**	r.g., i.p.
Gemma RR	895	$0.43^{\text{n.s.}}$	$1.43^{\text{n.s.}}$	.699	.129	verif.
Gemma SEQ	882	$0.44^{\text{n.s.}}$	$1.45^{\text{n.s.}}$	.693	.155	verif.
Gemma SEL	861	$0.60^{\text{n.s.}}$	14.8^{***}	$<.001$	.330	r.g., v., coord.
Llama RR	868	3.81^{**}	14.8^{**}	.002	.296	coord.
Llama SEQ	870	4.45^{**}	13.7^{**}	.003	.204	coord.
Llama SEL	865	17.3^{***}	22.5^{***}	$<.001$	.126	i.p., r.g.

Key observations. (1) **Seven of nine configurations** show concordant OLS and ME conclusions: Qwen RR/SEQ remain strongly function-driven (LRT $\chi^2 > 150$); Gemma RR/SEQ remain non-significant; Llama remains significant across all topologies though with negligible effect sizes (marginal $R^2 < 0.03$). (2) **Qwen SEL flips to non-significant** under ME (LRT $\chi^2 = 1.45$, $p = .229$). The ICC of 0.726 indicates that 73% of variance in remove-ablation similarity is between-task. With only two dominant function types (ACKNOWLEDGMENT and VERIFICATION) remaining after $n \geq 30$ filtering, the OLS effect reflects task-level confounding rather than within-task functional differentiation. (3) **Gemma SEL flips to significant** under ME (LRT $\chi^2 = 14.8$, $p < .001$, ICC $= 0.33$). However, this comparison involves only two function types (ACKNOWLEDGMENT and INFORMATION_PROVISION) after excluding three low-frequency categories, limiting interpretability. (4) Both flips occur under selector topology, which introduces strong task-level clustering (ICC > 0.3 for 4/9 configurations vs. only Qwen under RR/SEQ). This suggests that dynamic routing creates task-dependent communication patterns that violate the OLS independence assumption.

K Mistral-7B Cross-Backbone Validation

To test whether communication modes generalize beyond our three primary backbones, we ran the full pipeline on Mistral-7B-Instruct-v0.3 across all three topologies (300 GSM8K tasks each, same seed). Mistral uses the ChatML template with an explicit `<|im_end|>` stop token to prevent EOS residue in multi-turn conversations.

Accuracy. Mistral achieves 52% accuracy across all three topologies (RR: 52%, SEQ: 52%, SEL: 52%), substantially below Qwen (90%) and Gemma (80%) but comparable to Llama's selector performance (56%).

Function type distribution. Mistral communication is dominated by coordination and acknowledgment ($\sim$90% combined), with minimal information provision ($< 3\%$). Under selector topology, acknowledgment surges to 67%.

Table 11: Mistral-7B function type distribution (%) by topology.

Function type	RR	SEQ	SEL
COORDINATION	55.5	50.7	25.4
ACKNOWLEDGMENT	34.9	41.7	66.9
REASONING_GUIDANCE	6.8	6.2	6.3
INFORMATION_PROVISION	2.7	1.4	0.7

Communication mode: weakly differentiated. Table 12 summarizes the statistical analysis. Only the round-robin topology produces a significant function type effect (OLS $F = 6.6$, $p < .001$, $R^2 = 0.16$). Sequential and selector topologies show weaker or non-significant effects. Cross-model annotation agreement is fair ($\kappa \approx 0.34$), below the primary backbones, consistent with Mistral's less distinct functional boundaries.

Table 12: Mistral-7B statistical summary across topologies.

Topology	KW H	OLS F	R^2	κ
RR	22.4^{***}	6.6^{***}	.158	.344
SEQ	7.6	37.6^{***}	.056	.353
SEL	5.2	1.5	.065	.315

Interpretation. Mistral occupies an intermediate position between Llama's undifferentiated mode and Qwen's function-driven mode. The structured turn-taking of round-robin activates a degree of functional differentiation that disappears under less constrained topologies. This *topology-sensitive* pattern—where communication mode depends on external structure rather than being an intrinsic model property—extends the three-mode taxonomy and suggests that communication mode is jointly determined by model capability and conversational structure.

L DAMSL/ISO Taxonomy Mapping

Our five categories map onto established dialogue act frameworks as follows. INFORMATION PROVISION corresponds to DAMSL *Assert* Core and Allen [1997] and ISO 24617-2 *Inform* Bunt et al. [2012]. REASONING GUIDANCE maps to DAMSL *Action-directive* / ISO *Instruct*. VERIFICATION aligns with DAMSL *Agreement/Disagreement* and ISO *Confirm/Disconfirm* when evaluative, or *Correction* when error-correcting. COORDINATION subsumes FIPA *request*, *propose*, and *assign* Foundation for Intelligent Physical Agents [2002]. ACKNOWLEDGMENT corresponds to ISO *Auto-positive* feedback.

M Replacement Text Sensitivity

Our neutralize ablation uses "Acknowledged."—itself an acknowledgment-type utterance—as the replacement text. To test sensitivity to this choice, we reran neutralize ablation on 50 Qwen RR tasks with three placeholders: "Acknowledged.", "[message removed]", and "This message has been redacted." Table 13 reports per-function-type mean semantic similarity under each placeholder.

Table 13: Neutralize ablation: mean semantic similarity by function type and replacement text (Qwen RR, 50 tasks, 150 messages). Lower similarity indicates greater impact from neutralization.

Function Type	**Acknowledged.**	**[removed]**	**Redacted**
INFORMATION_PROVISION	0.146	0.107	0.050
REASONING_GUIDANCE	0.174	0.133	0.079
VERIFICATION	0.156	0.162	0.083
COORDINATION	0.352	0.319	0.181
ACKNOWLEDGMENT	0.134	0.148	0.051

Ranking stability. Pairwise Spearman rank correlations across the five function types are: "Acknowledged." vs. "[removed]" $\rho = 0.50$, "Acknowledged." vs. "Redacted" $\rho = 0.80$, "[removed]" vs. "Redacted" $\rho = 0.90$ ($p = .037$). The mean $\rho = 0.73$ indicates moderate-to-strong ranking stability. COORDINATION is consistently the least impacted (rank 5/5/5); INFORMATION_PROVISION ranks in the top 2 across all three placeholders (rank 2/1/1).

Acknowledged-specific confound. Under the "Acknowledged." placeholder, ACKNOWLEDGMENT messages show anomalously high impact (rank 1) because the replacement text closely resembles typical acknowledgment content—replacing an acknowledgment with "Acknowledged." preserves semantic similarity less than expected, as the regenerated next message responds to near-identical content. Excluding this confounded condition, the "[removed]" and "Redacted" placeholders agree strongly ($\rho = 0.90$), confirming that the neutralize gradient is robust to placeholder choice. Crucially, the remove ablation—which produces our primary finding (the counter-intuitive impact gradient)—is unaffected by replacement text choice, as it deletes messages entirely.

N Capability Confound Analysis

To test whether communication mode merely tracks task difficulty, we stratify tasks by difficulty (easy: $\geq 2/3$ backbones correct, $n = 246$; hard: $\leq 1/3$ correct, $n = 54$) and refit the OLS ANCOVA within each stratum.

Table 14: OLS ANCOVA by task difficulty stratum. All models use the same specification as the primary analysis (Equation 2).

Backbone	**Difficulty**	n	F	p	R^2
Qwen	Easy	735	196.6^{***}	$<.001$	.456
Qwen	Hard	160	37.3^{***}	$<.001$	.444
Gemma	Easy	726	$0.54^{\text{n.s.}}$	.657	.354
Gemma	Hard	152	$0.73^{\text{n.s.}}$	.483	.286
Llama	Easy	717	4.17^{**}	.006	.025
Llama	Hard	144	4.49^{*}	.013	.074

Qwen's function-driven mode persists on hard tasks ($R^2 = 0.444$), virtually unchanged from easy tasks ($R^2 = 0.456$). Gemma remains position-driven (function type non-significant for both strata), and Llama remains undifferentiated ($R^2 < 0.08$). The three modes are backbone-intrinsic properties rather than artifacts of task-relative capability.

NeurIPS Paper Checklist

1. **Claims** [Yes]All claims are supported by empirical evidence from our experiments. Ablation methodology is described as "perturbation-based" to accurately reflect its non-causal nature.
2. **Limitations** [Yes]Section 7 discusses limitations including single task domain, single scale, LLM-based annotation circularity, self-annotation bias, and greedy decoding.
3. **Theory Assumptions and Proofs** [N/A]Our work is empirical; no theoretical proofs are presented.
4. **Experimental Result Reproducibility** [Yes]Section 4–5 specify all models, hyperparameters, framework versions, and statistical tests. Greedy decoding ensures deterministic outputs. Code and data will be released upon acceptance.
5. **Open access to data and code** [Yes]Code, traces, and annotations will be released upon acceptance under an open-source license. GSM8K is publicly available.
6. **Experimental Setting/Details** [Yes]Sections 4–5 provide full experimental details including model names, parameter counts, generation settings, annotation protocol, and statistical framework.
7. **Compute Resources** [Yes]Experiments were conducted on 4×RTX 6000 Pro (96GB each) GPUs. Total compute is modest: all 9 configurations complete in approximately 24 hours.
8. **Code Of Ethics** [Yes]This research studies existing open-weight models on a public benchmark. No new models are trained. No human subjects are involved.
9. **Broader Impacts** [Yes]Our work characterizes communication in multi-agent systems to improve efficiency and interpretability. We do not foresee negative societal impacts beyond those inherent to LLM deployment generally.
10. **Safeguards** [N/A]Our work analyzes existing model behaviors on mathematical reasoning tasks and does not introduce new risks.
11. **Licenses** [Yes]GSM8K is MIT-licensed. All three model backbones use permissive open-weight licenses. AG2/AutoGen is Apache 2.0.
12. **New Assets** [Yes]We release interaction traces, function type annotations, and ablation results. These will be documented and released under an open license upon acceptance.
13. **Crowdsourcing and Research with Human Subjects** [N/A]No human subjects or crowdsourcing were involved.
14. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects** [N/A]No human subjects research.